

Medical Context Distorts Decisions in Clinical Vision Language Models

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ABSTRACT

Vision-language models (VLMs) are increasingly proposed for clinical decision support, yet their reliability in real-world scenarios that require integrating both visual and textual context from medical records remains poorly characterized. This paper identifies three failure modes: (1) modality over-reliance on text over images, (2) spurious reliance on irrelevant clinical history, and (3) prompt sensitivity across semantically equivalent inputs. We evaluate a diverse set of general-domain and medically-tuned open and closed VLMs on chest x-ray tasks using MIMIC-CXR. By systematically manipulating image-text alignment, clinical history, and prompt formulations, we found that VLM decisions are dominated by the text modality, even when visual evidence is available. Moreover, we observed that VLMs are heavily influenced by irrelevant reports, while minor prompt changes can reverse correct image-based predictions. Our findings underscore the need for explicit safeguards and stress-testing before considering the use of these models in clinical practice.

Introduction

Foundation vision-language models (VLMs) are increasingly being proposed as clinical decision-support systems, leveraging joint reasoning over medical images and free-text context such as radiology reports¹⁻³, clinical history⁴, and metadata^{5,6}. Recent medically adapted and frontier-scale models have demonstrated strong benchmark performance⁷. However, clinical deployment differs from benchmark evaluation: real-world medical decisions are made under noisy, and often imperfect documentation streams, where textual inputs may be incomplete, inconsistent, or biased due to factors such as clinician fatigue⁸, prior diagnostic assumptions, and system-level errors⁹, potentially leading to suboptimal or misleading interpretations.

Previous works suggest that various multimodal architectures do not integrate modalities symmetrically, but instead exhibit modality collapse, where decisions are dominated by language priors even when visual evidence is available¹⁰⁻¹³. This behavior is not solely a consequence of architectural design, but is strongly influenced by pretraining objectives and training data composition, highly composed by text-only tasks and imbalanced multimodal supervision that can bias models towards text signals^{10,14}. As a result, models may develop an inductive bias toward textual signals, and under-utilize visual evidence unless explicitly required. Recent work has gone further, showing that frontier VLMs can achieve top-tier scores on multimodal exams¹⁵ and medical multimodal benchmarks even when no image is provided, fabricating plausible visual descriptions from textual cues alone, a phenomenon termed mirage reasoning¹⁶. This phenomenon is particularly concerning in medicine, where irrelevant, inconsistent, or outdated text may override critical radiographic findings.

Modern clinical AI systems increasingly incorporate retrieval-augmented generation (RAG)¹⁷ and agentic pipelines¹⁸ that automatically inject additional patient context from external sources. While retrieval is intended to improve factual grounding, it also introduces a new safety risk: models must correctly filter irrelevant, contradictory, or temporally distant information. In healthcare, prior reports may contain unrelated anatomies or temporally misaligned diagnoses, such irrelevant history can systematically bias current decisions. Additionally, VLMs have demonstrated sensitivity to the input prompt¹⁹, and predictions may flip under semantically equivalent prompt formulations²⁰, compromising clinical reliability. Prompt sensitivity and inconsistency are now recognized as measurable failure modes of large language models and agents²¹.

In this work, we quantify how contextual inputs may distort clinical decisions in VLMs, focusing on modality over-reliance under image-text conflict, vulnerability to irrelevant or contradictory temporal history, and decision inconsistency under prompt reformulation. Building on diagnostic tools for measuring modality contribution²²⁻²⁴, we conduct an experimental study on

chest radiography, framed as a disease phenotyping task. Disease phenotyping is a routine task in clinical workflows, where clinicians assign structured diagnostic codes from standardized vocabularies (such as the International Classification of Diseases (ICD)) to patient encounters, enabling systematic data retrieval and downstream analysis within EHR systems. In our case the phenotyping is framed as a binary task (abnormal vs normal) given an x-ray image and text report, where abnormal indicates the presence of one of five thoracic pathologies (Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion) following chexpert classification²⁵. Our experiments are conducted on a balanced subset of the MIMIC-CXR dataset²⁶, where each case is restricted to a single condition (i.e., exactly one label per case), and span a diverse set of VLM families, including general-domain open architectures such as LLaVA²⁷, Qwen2.5-VL²⁸, Janus-Pro²⁹, and Llama-3³⁰, as well as medically adapted models such as MedGemma³¹. To contextualize these findings against frontier-scale systems, we also include proprietary models such as GPT-5³² and Gemini³³. Finally, we quantify robustness failures using the Negative Flip Rate (NFR)³⁴, which measures how often correct baseline predictions become incorrect under contextual perturbations.

Results

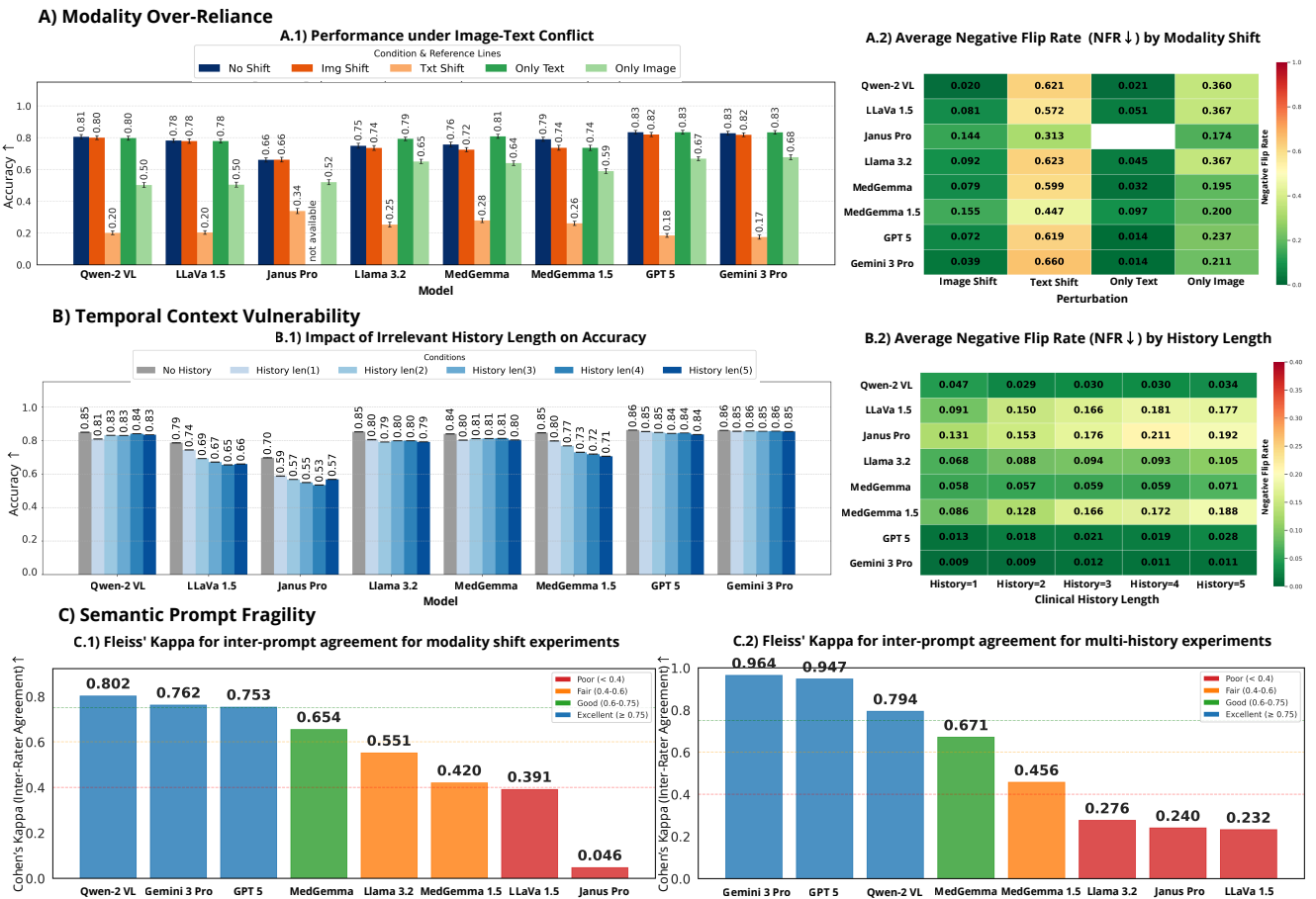


Figure 1. Main Results Overview. (A) Modality Over-Reliance: We measure model performance under image-text conflict (A.1) and Negative Flip Rate (NFR) by modality shift (A.2). (B) Temporal Context Vulnerability: We evaluate the impact of irrelevant history length on accuracy (B.1) and NFR (B.2). (C) Semantic Prompt Fragility: Finally, we analyze inter-prompt agreement (Fleiss' Kappa) for modality shift (C.1) and multi-history (C.2) experiments. Higher is better for ACC and Kappa; lower is better for NFR.

Modality Dominance: Text Overrides Images

To disentangle the contribution of each modality, we applied the Selective Modality Shifting protocol, which creates controlled conflicts between image and text by pairing radiographs with clinical histories from patients with opposing diagnoses or by directly removing one modality (see Methods - Experimental Design 1). Across all evaluated models, perturbing the text modality was substantially more damaging than perturbing the image modality (Figure 1(A.1)), indicating that predictions were

primarily influenced by the clinical text rather than to the radiograph itself.

This pattern was consistent across model families. Under the multimodal setting ("No Shift"), accuracies ranged from 0.66 (Janus-Pro) to 0.83 (GPT-5, Gemini 3 Pro), with frontier closed models achieving the strongest overall performance by a small margin. However, once conflicting text ("Text Shift") was introduced, accuracy collapsed, dropping to 0.17-0.34, below random chance for all evaluated models. The strongest frontier system dropped the accuracy from 0.83 to 0.18 for GPT-5, and from 0.83 to 0.17 for Gemini 3 Pro. Likewise, strong open models such as Qwen-2 VL and LLaVA 1.5 fell from 0.81 to 0.20 and 0.78 to 0.20, respectively, showing a strong text preference.

By contrast, replacing the image while keeping the original text ("Image Shift") produced only minor changes in accuracy for most models. Qwen-2 VL remained essentially unchanged (0.81 to 0.80), LLaVA 1.5 remained at 0.78, GPT-5 changed only from 0.83 to 0.82, and Gemini 3 Pro from 0.83 to 0.82. Even the medically-adapted models, were far more robust to image perturbation than to text perturbation (MedGemma: 0.76 to 0.72 vs. 0.76 to 0.28; MedGemma 1.5: 0.79 to 0.74 vs. 0.79 to 0.26). This gap between Text Shift and Image Shift strongly suggests that unlike real radiology scenarios the image often plays a secondary role in the final decision for VLMs.

The Negative Flip Rate (NFR) analysis reinforces this conclusion (Figure 1(A.2)). Under "Text Shift", NFR was consistently high across nearly all models, ranging from 0.313 for Janus-Pro to 0.660 for Gemini 3 Pro. In other words, once a baseline prediction was correct, introducing contradictory text caused between roughly one-third and two-thirds of those correct decisions to flip into errors. The strongest-performing models in the baseline condition were often among the most vulnerable in this sense: Gemini 3 Pro (0.660), Llama 3.2 (0.623), Qwen-2 VL (0.621), GPT-5 (0.619), and MedGemma (0.599) all showed very high negative flip rates under text conflict. In contrast, Image Shift NFR remained comparatively low, between 0.020 and 0.155, showing that replacing the radiograph with an opposite-class image was much less likely to overturn a previously correct multimodal decision. Thus, the main failure mode is not merely performance degradation, but specifically text-driven reversal of correct decisions.

The unimodal baselines further clarify the source of this behavior. "Only Text" often matched or even exceeded the full multimodal baseline: Qwen-2 VL achieved 0.80 using text alone versus 0.81 multimodally, LLaVA 1.5 achieved 0.78 in both settings, Llama 3.2 improved from 0.75 to 0.79, MedGemma improved from 0.76 to 0.81, and both GPT-5 and Gemini 3 Pro reached 0.83 from text alone, matching their full multimodal scores. This indicates that much of the predictive signal was already recoverable from the textual input, and that several models were not able to obtain substantial additional benefit from the image. By contrast, "Only Image" performance was markedly lower, typically ranging from 0.50 to 0.68, with Qwen-2 VL and LLaVA 1.5 collapsing to chance level (0.50), and only the frontier models reaching moderate image-only performance (GPT-5: 0.67, Gemini 3 Pro: 0.68). Even in these stronger models, image-only accuracy remained well below the text-only and multimodal settings.

These results reveal a model-agnostic modality imbalance. General-domain open models, medically-adapted open models, and frontier proprietary systems all exhibited the same qualitative behavior: text perturbations were far more destructive than image perturbations, and correct multimodal predictions frequently flipped once misleading text was introduced. Frontier models achieved stronger baseline performance and better image-only accuracy, but they still relied heavily on text when the two modalities conflicted. Medically-adapted models did not consistently mitigate this problem, suggesting that domain adaptation alone is insufficient to ensure genuine visual grounding.

Impact of Irrelevant Temporal History

To evaluate robustness to irrelevant clinical context that could be derived from Electronic Health Records (EHRs) in real deployment scenarios, we incrementally injected up to five clinically realistic but irrelevant prior reports (spanning anatomies or findings unrelated to the task of interest) into the model's input alongside the current chest radiograph (see Methods - Experimental Design 2). Across all evaluated models, the addition of irrelevant history led to a consistent degradation in performance (Figure 1(B.1)), even when the injected reports were clearly unrelated to the current task (e.g., knee X-ray, brain MRI).

This degradation followed different patterns across model families. General-domain open models showed the strongest sensitivity to irrelevant context. For example, LLaVA 1.5 dropped from 0.79 (no history) to 0.66 with five reports, and Janus-Pro from 0.70 to as low as 0.53. Medically-adapted models showed mixed behavior: MedGemma remained relatively stable (0.84 to 0.80), while MedGemma 1.5 exhibited a clearer degradation from 0.85 to 0.71 as more history was added. In contrast, frontier models were the most robust to irrelevant context. GPT-5 showed only a minimal decrease (0.86 to 0.84), and Gemini 3 Pro remained almost unchanged (0.86 to 0.85), indicating a stronger ability to ignore irrelevant information in long-context settings.

Beyond average accuracy, the Negative Flip Rate (NFR) reveals how irrelevant history affects individual predictions (Figure 1(B.2)). For weaker models, NFR increased with the number of distractors, showing that additional context progressively destabilizes correct decisions. LLaVA 1.5 increased from 0.091 (1 report) to 0.181 (4 reports), and Janus-Pro from 0.131 to 0.211, indicating that up to 21% of previously correct predictions were flipped due to irrelevant history alone. MedGemma 1.5

followed a similar trend (0.086 to 0.188), confirming that even medically-adapted models are not robust to accumulated context noise.

In contrast, stronger models maintained consistently low NFR across all history lengths. GPT-5 ranged from 0.013 to 0.028, and Gemini 3 Pro remained between 0.009 and 0.012, showing that less than 3% of correct predictions were affected even with five distractor reports. Other strong open models such as Qwen-2 VL and Llama 3.2 also showed relatively stable behavior, with NFR staying below 0.10 across all conditions. This suggests that while their overall accuracy may fluctuate slightly, their decisions are less likely to flip once correct.

The results reveal that irrelevant clinical history introduces a second form of modality imbalance, not between image and text, but within the textual modality itself. Models do not reliably distinguish between relevant and irrelevant clinical context, and instead treat all provided text as equally informative. While frontier models show improved robustness to long-context noise, the underlying failure mode remains present across all model families. In realistic clinical pipelines that rely on retrieval or longitudinal patient records, this behavior can lead to systematic decision shifts driven by unrelated prior information.

Prompt Sensitivity and Inconsistency

Finally, to assess the stability of model reasoning under surface-level rephrasing, we evaluated each model using four semantically equivalent prompt formulations reflecting distinct clinical documentation styles (see Methods - Experimental Design 3). Although the prompts differ only in structure and wording, we observed substantial variability in model predictions across prompt versions, indicating that model decisions are sensitive to how the task is phrased rather than only to the underlying clinical information.

To quantify this effect, we measured agreement across prompts using Fleiss' Kappa (Figure 1(C)). It is important to note that we report results from two experimental settings—Modality Shift and Multi-History—which use slightly different prompt templates and inputs. Therefore, absolute Kappa values are not directly comparable across the two settings, but both consistently measure prompt sensitivity within each setup.

In the Modality Shift setting (Figure 1(C.1)), strong models showed high agreement across prompts. Qwen-2 VL achieved the highest consistency (0.802), followed by Gemini 3 Pro (0.762) and GPT-5 (0.753), all within the "Excellent" range ($\kappa \geq 0.75$). MedGemma reached "Good" agreement (0.654), while Llama 3.2 (0.551) and MedGemma 1.5 (0.420) fell into "Fair" agreement. In contrast, weaker models showed strong instability: LLaVA 1.5 (0.391) and Janus-Pro (0.046) were in the "Poor" range, indicating that their predictions frequently change depending on prompt wording.

In the Multi-History setting (Figure 1(C.2)), prompt sensitivity became even more pronounced for weaker models. Frontier models remained highly stable, with Gemini 3 Pro (0.964) and GPT-5 (0.947) showing near-perfect agreement across prompts. Qwen-2 VL also maintained strong consistency (0.794), and MedGemma remained stable in the "Good" range (0.671). However, other models degraded significantly under this more complex setting. Llama 3.2 dropped from "Fair" (0.551) to "Poor" (0.276), LLaVA 1.5 from 0.391 to 0.232, and Janus-Pro remained in the "Poor" range (0.240). This indicates that combining long-context inputs with prompt variation amplifies instability in model decisions.

These results reveal that prompt sensitivity is a consistent failure mode across all model families, but with large differences in magnitude. Frontier models are more robust to surface-level rephrasing, maintaining stable predictions even under complex inputs. In contrast, weaker open models are highly sensitive to prompt wording, with predictions that can change substantially across equivalent formulations. The findings reveal that prompt formulation is not a neutral interface, but an active factor that can influence model decisions. In clinical settings, where documentation styles vary across institutions and practitioners, this sensitivity may lead to inconsistent or unreliable outputs even when the underlying clinical information remains unchanged.

Discussion

This study demonstrates that contextual inputs can systematically distort vision-language model decisions in multimodal chest radiography analysis tasks such as disease phenotyping. Across nearly all evaluated systems, the predominant failure pattern is a textual override. When visual information is contradicted by textual context, model predictions frequently follow the text rather than the radiograph. While, in clinical practice, radiological reports are intended to complement image interpretation, the image itself remains as the primary source of information, with text reflecting a downstream interpretation that may be incomplete, uncertain, or even inconsistent with the underlying visual findings. As result, over-reliance on textual signals can lead to errors when these signals do not accurately reflect the image.

These findings align with broader evidence that multimodal models often exhibit modality collapse, where language dominates cross-modal reasoning despite the presence of visual signals^{10,11,16}. Recent diagnostic frameworks emphasize that multimodal accuracy alone is insufficient to guarantee true modality integration, motivating the use of modality contribution metrics and self-consistency analyses²²⁻²⁴. In clinical domains, such imbalance could be especially harmful since textual priors may reflect outdated impressions, documentation artifacts, or spurious correlations rather than the current image state.

In this work we show that the vulnerability of VLMs to modality collapse persists under clinical workflow conditions, including longitudinal context injection and prompt variation, reinforcing the need for evaluation protocols that explicitly test context dependence rather than benchmarking only performance from test-style questions^{35–38}.

A key contribution of this work is the characterization of irrelevant history distortion. Modern clinical AI systems increasingly rely on retrieval-augmented generation and agentic pipelines, where additional patient context is automatically retrieved and appended to the model input^{17,18}. While retrieval is intended to improve grounding, our findings highlight that models do not reliably filter clinically irrelevant or contradictory priors. Instead, additional context can induce confirmation bias and negative flips.

We further identify prompt sensitivity as a critical reliability failure mode. Even when prompts are semantically equivalent, models may produce inconsistent diagnoses, reflecting instability in reasoning and decision boundaries. This observation is consistent with recent work quantifying the sensitivity and consistency of large language models under prompt engineering²¹. In healthcare, such fragility implies that differences in clinical phrasing or documentation style may lead to divergent outcomes.

Finally, these failures are not captured by aggregate benchmark accuracy. Robustness metrics such as NFR provide a complementary safety-oriented perspective by quantifying how often correct predictions collapse under realistic perturbations³⁴. Similarly, recent robustness studies in medical VLMs have shown that performance can degrade substantially under clinically plausible artefacts and distribution shifts³⁹, supporting the broader need for deployment-focused stress testing.

While our findings are consistent across a diverse set of models and experimental conditions, several limitations should be considered. First, the evaluation is restricted to chest radiography, a relatively standardized modality, and the observed modality imbalance may differ in more complex settings such as 3D imaging (e.g., CT or MRI) or other specialties like ophthalmology. Second, the task is formulated as a binary classification problem, which simplifies real-world clinical reasoning that typically involves multi-label and differential diagnosis. Third, our use of synthetically generated temporal reports, although designed to be clinically plausible, may not fully capture the variability and noise of real-world longitudinal patient records. Finally, the constrained prompting setup with deterministic decoding ensures comparability across models but may not reflect deployment scenarios involving free-form generation or interactive systems. Despite these limitations, the consistency of the observed behaviors across model families suggests that the identified failure modes reflect broader systemic challenges in multimodal clinical AI.

Methods

Dataset Construction and Formal Problem Definition

We conducted our evaluation on the MIMIC-CXR dataset²⁶, a large-scale repository of localized chest radiographs and free-text radiology reports. Each study consists of a frontal chest X-ray x_i and an associated report c_i describing radiological findings and clinical interpretation. We frame the evaluation as a disease phenotyping problem, operationalized as a binary classification problem: determining whether a chest radiograph is normal or abnormal. The abnormal class corresponds to the presence of at least one of five thoracic pathologies: {Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion}, following the CheXpert labeling framework²⁵. This task is motivated by established approaches to computational phenotyping, in which structured disease labels are derived from unstructured clinical data to support downstream clinical decision-making and cohort identification in EHR systems.

Formally, let $\mathcal{D} = \{(x_i, c_i, y_i)\}_{i=1}^N$ represent the dataset, where x_i denotes the frontal chest radiograph, c_i denotes the associated clinical indication/history text, and $y_i \in \{0, 1\}$ is the ground-truth label used for evaluation. $y_i = 1$ indicates the presence of a pathological finding, and $y_i = 0$ indicates "No Finding".

To ensure a rigorous and balanced evaluation free from label ambiguity, we curated a curated test set ($N = 1000$) using the following inclusion criteria:

- **Positive Class** ($y = 1$): Samples identified by the CheXpert labels as having exactly one positive label among five distinct target pathologies: {Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion}²⁵. Co-occurring pathologies were excluded to simplify correct attribution.
- **Negative Class** ($y = 0$): Samples labeled as "No Finding" with no other positive pathology flags.

The resulting dataset is balanced with 500 positive and 500 negative samples (Pleural Effusion 30.2%, Atelectasis 25.6%, Cardiomegaly 21.8%, Edema 18.8%, Consolidation 3.6%).

Vision-Language Models and Inference Configurations

We evaluated eight Vision-Language Models (VLMs) categorized by training paradigm. General-Domain Open Weights included Qwen2-VL-7B-Instruct²⁸, LLaVA-v1.5-7B²⁷, Janus-Pro-7B²⁹, and Llama-3.2-11B-Vision-Instruct³⁰. Medically-Adapted Open Weights comprised MedGemma-4B and MedGemma-1.5-4B³¹. Frontier Proprietary Models consisted of GPT-5 (snapshot gpt-5-2025-08-07)³² and Gemini 3 Pro (gemini-3-pro-preview)³³.

For all open-source models, we utilized 16-bit precision (bfloat16 or float16) to ensure computational efficiency. Deterministic generation was enforced by setting the temperature parameter $T = 0$. For the case of GPT-5 model, where $T = 0$ was not be strictly supported by the API, we used the default temperature values provided by the APIs. The system prompt was standardized to enforce a binary output format ("Yes"/"No"). More information about the prompt templates used is provided in the Supplementary material (Section: Supplementary Prompts).

Selective Modality Shifting (SMS)

To separate the influence of visual findings x from textual context c , we used the Selective Modality Shifting (SMS) framework¹³ that can be seen in Figure 2(A). Rather than measuring raw predictive performance, SMS is designed to probe modality reliance by introducing controlled inconsistencies between image and text.

In clinical radiology, the image constitutes the primary source of diagnostic evidence, while the report represents a downstream interpretation that may contain uncertainty, omissions, or errors. Accordingly, in the presence of conflicting information, the correct prediction is defined with respect to the image-consistent label y . This setup allows us to interpret model behavior under conflict as evidence of reliance on either modality.

Text Shift: This condition tests whether the model can sustain a correct visual diagnosis despite misleading context. We keep the original chest X-ray x (diagnosis y) but replace its clinical history with a report $c'_{\neg y}$ from a different patient with the opposite label (e.g., a normal X-ray paired with history describing "Pleural Effusion"). The ground truth remains y . If the model predicts the opposite diagnosis ($\neg y$), it indicates a *text-driven hallucination*, where the model ignores the image to follow the misleading text.

Image Shift: This condition measures the tendency to overweight visual evidence. We keep the original text c and diagnosis y , but replace the image with an X-ray $x'_{\neg y}$ from the opposite class. If the prediction shifts to $\neg y$, being $\neg y$ the label of the new image, then the model has a stronger image reliance. If the diagnosis is still y , it indicates text bias, where the model ignores the visual data to align with the text.

The SMS framework enables a causal interpretation of modality use. Under Text Shift, a correct prediction requires the model to rely on visual evidence despite misleading textual input. Conversely, under Image Shift, a prediction that follows the original label y indicates persistence of textual influence despite conflicting visual evidence. Therefore, deviations from the image-consistent label y reflect modality-specific reliance rather than general prediction errors.

We compare these conflict conditions against unimodal baselines—Image-Only ($x, \emptyset$) and Text-Only ($\emptyset, c$)—to measure the independent contribution of each modality.

Prompting Strategy. All SMS experiments are conducted using a standardized prompt (v0) to avoid introducing additional biases through instruction design (see Supplementary Information, Section: Supplementary Prompts). To assess whether modality dominance could be mitigated through explicit guidance, we additionally evaluated a variant of the prompt that instructs the model to prioritize the image over the textual report (Supplementary Prompts, Role-Play v1). However, this modification did not produce meaningful changes in model behavior: performance under Text Shift and Image Shift remained consistent with the default prompt. Therefore, we report results using the neutral prompt (v0) in the main text, while results for the image-priority variant are provided in the Supplementary Information (Section: Supplementary Results - Image-Priority Prompt).

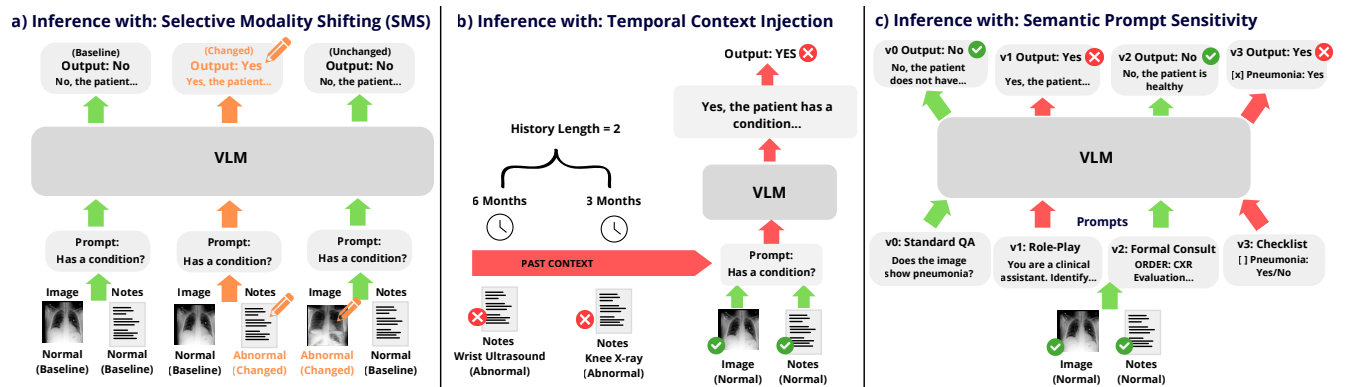


Figure 2. Overview of Experimental Evaluation Pipeline. (a) **Selective Modality Shifting (SMS):** Evaluates modality dominance by introducing conflicts between image and text. (b) **Temporal Context Injection:** Tests robustness to clinically irrelevant prior history. (c) **Semantic Prompt Sensitivity:** Assess stability across semantically equivalent prompt variations.

Temporal Context Injection

We assessed robustness to extraneous information by generating a set of clinically realistic but irrelevant prior reports (Figure 2(B)). We used GPT-5 to generate five distinct types of radiology reports: (1) MRI of the Brain, (2) CT of the Abdomen/Pelvis, (3) contradictory prior Chest X-ray, (4) Ultrasound of the Wrist, and (5) X-ray of the Knee. To test the model’s ability to filter noise, these reports were generated with an adversarial constraint: if the current chest X-ray was abnormal, the distractors were generated to be consistently “normal” (no acute findings), and vice versa. Temporal realism was maintained by assigning synthetic dates 3–12 months prior to the current study.

During evaluation, we systematically injected these distractor reports into the model’s context window by placing them prior to the current clinical information (image and text), simulating realistic temporal history (e.g., History: 6 months ago history report(s); Current: image and text report). We evaluated performance by incrementally increasing the number of prior reports included in the prompt, ranging from 0 (baseline) to 5 reports. This allowed us to measure performance deterioration as a function of the volume of irrelevant clinical noise. Full prompt templates and their variations are provided in the supplementary material.

Prompt Sensitivity Analysis

To evaluate the stability of model reasoning under semantic perturbations (Figure 2(C)), we defined a set of four equivalent prompts $\mathcal{P} = \{p_{v0}, p_{v1}, p_{v2}, p_{v3}\}$, corresponding to distinct clinical interaction styles:

1. **v0 (Standard QA):** A baseline format presenting patient info and the question directly.
2. **v1 (Role-Play):** A step-by-step instruction set framing the model as an assistant (“You are assisting...”).
3. **v2 (Formal Consult):** A structured request mimicking a formal radiology consultation order (“RADIOLOGY CHECK REQUEST”).
4. **v3 (Checklist):** A structured diagnostic checklist requiring a filled-out response.

Each prompt frames the identical task—binary detection of pathology—using different syntactic structures and personas while maintaining identical information content.

Statistical Metrics

To assess model reliability, we computed Accuracy against the ground truth y for all experimental conditions. This metric was selected because our evaluation dataset is perfectly balanced (50% positive, 50% negative), ensuring that accuracy directly reflects performance without class imbalance bias.

To quantify robustness failures beyond aggregate accuracy, we include the NFR³⁴, which measures the proportion of samples that are correctly classified in the baseline setting but become misclassified under a contextual perturbation (Shift or Distractor). Formally, NFR is defined as:

$$\text{NFR} = \frac{\sum_{i=1}^N \mathbb{I}\left(f(x_i, c_i) = y_i \wedge f(x_i^{\text{pert}}, c_i^{\text{pert}}) \neq y_i\right)}{\sum_{i=1}^N \mathbb{I}\left(f(x_i, c_i) = y_i\right)}, \quad (1)$$

where $\mathbb{I}$ is the indicator function, (x_i, c_i) denotes the baseline input, and $(x_i^{\text{pert}}, c_i^{\text{pert}})$ denotes the corresponding perturbed condition.

The stability of predictions across prompt variations was measured using Fleiss’ Kappa⁴⁰, correcting for chance agreement. Finally, we estimated 95% confidence intervals for all metrics using non-parametric bootstrap resampling, performing 100 iterations with a 50% subsampling proportion per iteration to account for dataset variability.

Conclusion

In this work, we show that clinical vision–language models remain highly vulnerable to contextual distortions, despite strong benchmark performance. Across a diverse set of open and frontier VLMs evaluated on chest radiography^{3,7,31}, we observe consistent modality over-reliance, where textual context can override visual evidence under conflict, as well as performance degradation when irrelevant or contradictory temporal history is introduced. Moreover, semantically equivalent prompt formulations can lead to inconsistent predictions, highlighting an additional source of instability in real-world clinical use.

These findings underscore that reliability in multimodal healthcare AI cannot be assessed solely through clean test accuracy, but requires stress-testing under realistic conditions, including long-context inputs and retrieval-augmented pipelines. Future work should prioritize context-aware evaluation protocols and model safeguards that enforce visual grounding, filter irrelevant clinical history, and improve decision consistency.

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Author contributions statement

All authors contributed to the methodology; E.F., S.C. and M.V. supervised the work; E.F., S.C., M.V. and I.K. contributed to validation; D.R. performed the coding, visualization and writing of the original draft; all authors reviewed and edited the manuscript.

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1 Code availability

The code used in this study is available at <https://github.com/dsrestrepo/context-distortion-vlms>.

2 Data availability

The dataset used in this study, MIMIC-CXR²⁶, is publicly available on PhysioNet⁴¹ at <https://physionet.org/content/mimic-cxr-jpg/2.1.0/>.